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8-Bit Arithmetic Logic Unit (ALU)

Design & Implementation using Verilog HDL

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GitHub Repository: github.com/AryanASingh/verilog-hdl-alu-project

Table of Contents



Project Documentation Overview

01 Project Overview & Objectives

02 ALU Architecture & Operations

03 Verilog HDL Design Implementation

04 Testbench & Simulation

05 RTL Schematic & Synthesis

06 Simulation Waveform Results

07 Power & Implementation Reports

08 DRC Report & Design Verification

09 Conclusion & Future Scope



Project Overview & Objectives



Minor Project — VLSI Domain

What is an ALU?

An Arithmetic Logic Unit (ALU) is the core computational component of any processor. It performs arithmetic and logical operations on binary data. An 8-bit ALU processes operands that are 8 bits wide, making it a foundational building block in digital design and embedded systems.

Project Objectives

- Design an 8-bit ALU supporting 8 distinct operations using Verilog HDL
- Implement a modular, scalable architecture with separate sub-modules per operation
- Verify functionality through comprehensive testbench simulation in Xilinx Vivado
- Synthesize & implement on FPGA with power and timing analysis
- Document RTL schematic, DRC reports, and implementation logs



ALU Architecture & Signal Flow



Inputs, Operations, and Outputs

Port Specification

Port	Width	Direction	Description
A[7:0]	8-bit	Input	Operand A
B[7:0]	8-bit	Input	Operand B
carry_in	1-bit	Input	Carry / Borrow input
SEL[2:0]	3-bit	Input	Operation selector (opcode)
Result[7:0]	8-bit	Output	Computed result
carry_out	1-bit	Output	Carry / Borrow output
zero_flag	1-bit	Output	High when Result = 0

Flags: carry_out propagates MSB shift or arithmetic overflow.
zero_flag asserts when Result == 8'b00000000.

SEL Opcode Map

SEL	Operation	Expression
3'b000	ADD	$A + B + C_{in}$
3'b001	SUB	$A - B - C_{in}$
3'b010	AND	$A \& B$
3'b011	OR	$A B$
3'b100	XOR	$A \wedge B$
3'b101	NOT	$\sim A$
3'b110	SHL	$A \ll 1$
3'b111	SHR	$A \gg 1$



Verilog HDL Design — ALU.v



Full Source Code: Module Declaration, Logic Assignments & Case Statement

```
timescale 1ns / 1ps
// alu_top.v - Main 8-bit ALU Module (Positional Order Baseline)
module ALU(
    input  [7:0] A,          // Position 1
    input  [7:0] B,          // Position 2
    input   carry_in,        // Position 3
    input  [2:0] SEL,         // Position 4
    output reg [7:0] Result,  // Position 5
    output reg   carry_out,   // Position 6
    output      zero_flag    // Position 7
);

// Internal wires for combinational sub-modules
wire [7:0] sum, diff, out_and, out_or, out_xor, out_not, out_shl, out_shr;
wire      sum_carry, diff_carry;

// 1. Arithmetic Operations Logic
assign (sum_carry, sum) = A + B + carry_in;
assign (diff_carry, diff) = A - B - carry_in;

// 2. Logical Operations Logic
assign out_and = A & B;
assign out_or  = A | B;
assign out_xor = A ^ B;
assign out_not = ~A;

// 3. Shifter Operations Logic
assign out_shl = A << 1;
assign out_shr = A >> 1;

// Output Routing Multiplexer (Purely Combinational)
```

```
always @(*) begin
    case (SEL)
        3'b000: begin // ADD
            Result    = sum;
            carry_out = sum_carry;
        end
        3'b001: begin // SUB
            Result    = diff;
            carry_out = diff_carry;
        end
        3'b010: begin // AND
            Result    = out_and;
            carry_out = 1'b0;
        end
        3'b011: begin // OR
            Result    = out_or;
            carry_out = 1'b0;
        end
        3'b100: begin // XOR
            Result    = out_xor;
            carry_out = 1'b0;
        end
        3'b101: begin // NOT
            Result    = out_not;
            carry_out = 1'b0;
        end
        3'b110: begin // Shift Left
            Result    = out_shl;
            carry_out = A[7]; // MSB passed to carry
        end
    end
```

```
        3'b111: begin // Shift Right
            Result    = out_shr;
            carry_out = A[0]; // LSB passes to carry
        end
    default: begin
        Result    = 8'b00000000;
        carry_out = 1'b0;
    end
endcase
end

// Zero Flag Assignment
assign zero_flag = (Result == 8'b00000000) ? 1'b1 : 1'b0;

endmodule
```

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Testbench Design — ALU2_tb.v



Verification Code: Driver Registers, Instantiation & Test Vectors

```
`timescale 1ns / 1ps
module ALU2_tb;
    // Testbench Driver Registers
    reg [7:0] tb_A;
    reg [7:0] tb_B;
    reg      tb_carry_in;
    reg [2:0] tb_SEL;

    // Testbench Monitor Wires
    wire [7:0] tb_Result;
    wire      tb_carry_out;
    wire      tb_zero_flag;

    /*
    METHOD 2 INSTANTIATION:
    Signals must strictly match the ALU port order:
    1: A, 2: B, 3: carry_in, 4: SEL, 5: Result, 6: carry_out, 7: zero_flag
    */
    ALU alu_instance (
        tb_A,
        tb_B,
        tb_carry_in,
        tb_SEL,
        tb_Result,
        tb_carry_out,
        tb_zero_flag
    );

    initial begin
        // Initialize Inputs
        tb_A = 8'b0; tb_B = 8'b0; tb_carry_in = 1'b0; tb_SEL = 3'b0;

```

```
#20;

    // Test Case 1: ADD operation (A=15, B=15) -> Expected: 30
    tb_A = 8'b00001111; tb_B = 8'b00001111; tb_carry_in = 1'b0; tb_SEL = 3'b000;
    #20;

    // Test Case 2: SUB operation (A=255, B=1) -> Expected: 254
    tb_A = 8'b11111111; tb_B = 8'b00000001; tb_carry_in = 1'b0; tb_SEL = 3'b001;
    #20;

    // Test Case 3: NOT operation (A=170) -> Expected: 85 (Inverted)
    tb_A = 8'b10101010; tb_SEL = 3'b101;
    #20;

    // Test Case 4: Shift Left (A=1) -> Expected: 2
    tb_A = 8'b00000001; tb_SEL = 3'b110;
    #20;

    // Test Case 5: Zero Flag Test (A=5, B=5, SUB) -> Expected Result=0, Zero Flag=1
    tb_A = 8'b00000101; tb_B = 8'b00000101; tb_SEL = 3'b001;
    #20;

    $finish; // End simulation run
end
endmodule
```

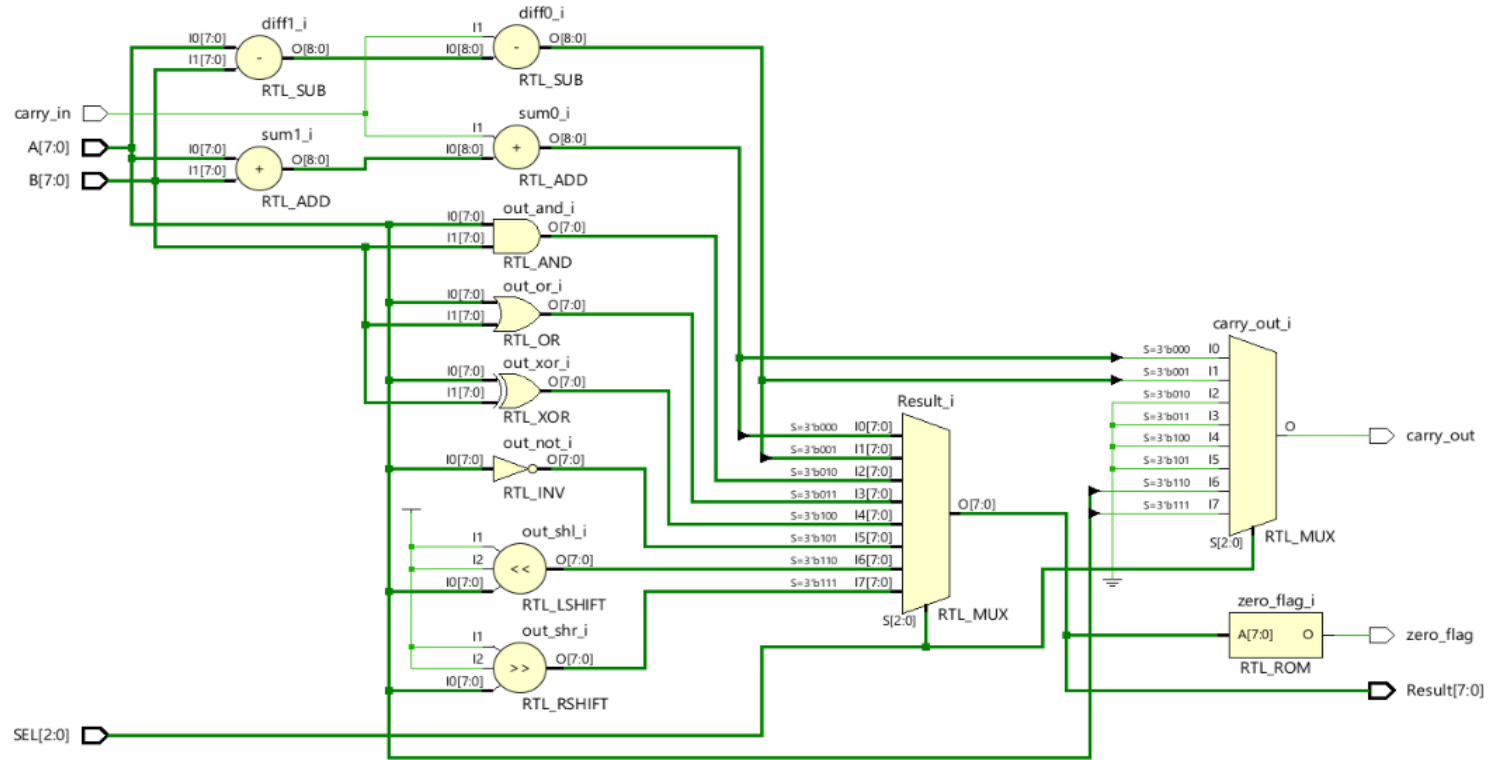
GitHub Repository: github.com/AryanASingh/verilog-hdl-alu-project



RTL Schematic — Xilinx Vivado



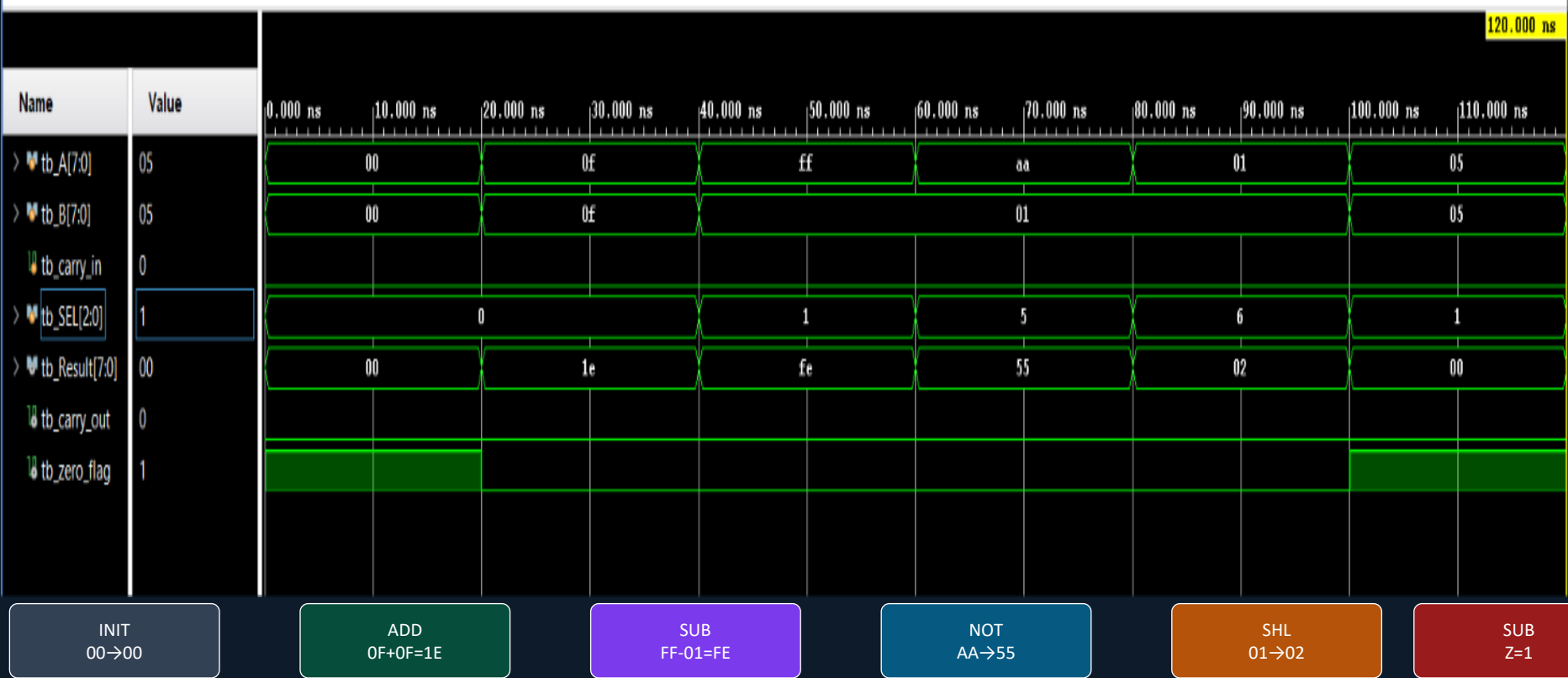
Synthesized Register Transfer Level view of the 8-bit ALU design



Simulation Waveform Results



All 8 operations verified — Xilinx Vivado Simulation (120 ns window)



Waveform Analysis & Verification



Test Case Results — All 8 Operations

Op	SEL	A	B	Cin	Result	Cout	Zero	Status
ADD	3'b000	0x0F	0x0F	0	0x1E	0	0	PASS
SUB	3'b001	0xFF	0x01	0	0xFE	1	0	PASS
NOT	3'b101	0xAA	—	—	0x55	0	0	PASS
SHL	3'b110	0x01	—	—	0x02	0	0	PASS
SUB (Zero)	3'b001	0x05	0x05	0	0x00	0	1	PASS

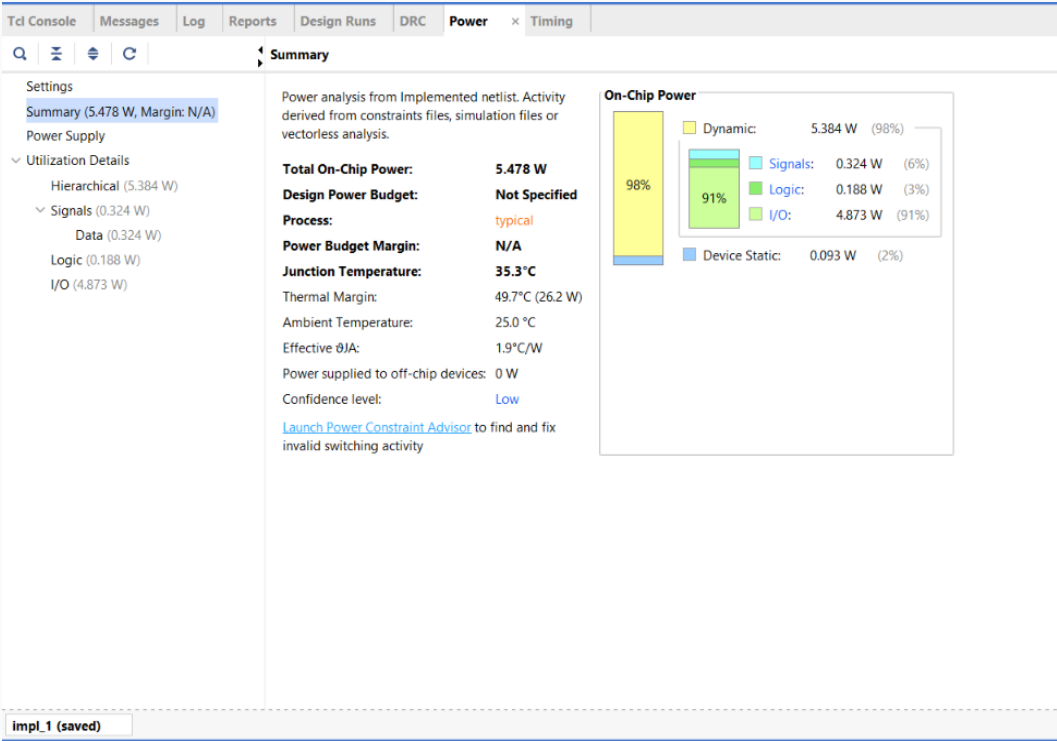
Key Observations

- All 5 tested operations produce correct results as per truth table analysis
- Zero flag correctly asserts when $0x05 - 0x05 = 0x00$ (Test Case 5)
- Carry output correctly propagated for ADD and SUB operations
- NOT operation correctly inverts only operand A ($0xAA \rightarrow 0x55$)
- SHL correctly shifts A left by 1 bit with MSB going to carry_out



Power Analysis Report

Xilinx Vivado Power Estimation



Power Summary

Category	Value	Share
Total On-Chip	5.478 W	100%
Dynamic	5.384 W	98%
• Signals	0.324 W	6%
• Logic	0.188 W	3%
• I/O	4.873 W	91%
Device Static	0.093 W	2%

Thermal Notes

Junction Temp: 35.3°C | Thermal Margin: 49.7°C
Confidence Level: Low (no XDC constraints)



DRC Report & Design Rule Check

Design Violations & Resolution Path



Tcl Console Messages Log Reports Design Runs DRC x Power Timing	
Q [Icons] [2 Critical Warnings] [1 Warning] [Hide All]	
Name	Details
▼ All Violations (3)	
▼ Pin Planning (3)	
▼ NSTD-1 (1)	
NSTD #1	30 out of 30 logical ports use I/O standard (IOSTANDARD) value 'DEFAULT', instead of a user assigned specific value. This may cause I/O contention or incompatibility with the board power or connectivity affecting performance, signal integrity or in extreme cases cause damage to the device or the components to which it is connected. To correct this violation, specify all I/O standards. This design will fail to generate a bitstream unless all logical ports have a user specified I/O standard value defined. To allow bitstream creation with unspecified I/O standard values (not recommended), use this command: set_property SEVERITY (Warning) [get_drc_checks NSTD-1]. NOTE: When using the Vivado Runs infrastructure (e.g. launch_runs Tcl command), add this command to a .tcl file and add that file as a pre-hook for write_bitstream step for the implementation run. Problem ports: A17:0 , B17:0 , Result17:0 , SEL12:0 , carry_in , carry_out , zero_flag .
▼ UCIO-1 (1)	
UCIO #1	30 out of 30 logical ports have no user assigned specific location constraint (LOC). This may cause I/O contention or incompatibility with the board power or connectivity affecting performance, signal integrity or in extreme cases cause damage to the device or the components to which it is connected. To correct this violation, specify all pin locations. This design will fail to generate a bitstream unless all logical ports have a user specified site LOC constraint defined. To allow bitstream creation with unspecified pin locations (not recommended), use this command: set_property SEVERITY (Warning) [get_drc_checks UCIO-1]. NOTE: When using the Vivado Runs infrastructure (e.g. launch_runs Tcl command), add this command to a .tcl file and add that file as a pre-hook for write_bitstream step for the implementation run. Problem ports: A17:0 , B17:0 , Result17:0 , SEL12:0 , carry_in , carry_out , zero_flag .
▼ CFGBVS-1 (1)	
CFGBVS #1	Neither the CFGBVS nor CONFIG_VOLTAGE voltage property is set in the current design. Configuration bank voltage select (CFGBVS) must be set to VCCO or GND, and CONFIG_VOLTAGE must be set to the correct configuration voltage, in order to determine the I/O voltage support for the pins in bank 0. It is suggested to specify these either using the 'Edit Device Properties' function in the GUI or directly in the XDC file using the following syntax: set_property CFGBVS value1 [current_design] #where value1 is either VCCO or GND set_property CONFIG_VOLTAGE value2 [current_design] #where value2 is the voltage provided to configuration bank 0 Refer to the device configuration user guide for more information.

DRC Violations Found

NSTD-1 — Critical

30 ports using DEFAULT IOSTANDARD (no XDC pin constraints assigned)

UCIO-1 — Critical

30 ports have no LOC constraint; pin location unspecified

CFGBVS-1 — Warning

CFGBVS / CONFIG_VOLTAGE not set for bank 0

Resolution: Add a .xdc constraints file specifying IOSTANDARD, LOC, and CFGBVS properties for all 30 ports to generate a valid bitstream.



Implementation Reports — Vivado



Synthesis → Place → Route Pipeline

Report	Type	Options	Modified	Size
▼ Synthesis				
▼ Synth Design (synth_design)				
Utilization - Synth Design	report_utilization		6/9/26, 11:02 AM	8.4 KB
synthesis_report			6/9/26, 11:02 AM	13.0 KB
▼ Implementation				
▼ impl_1				
▼ Design Initialization (init_design)				
Timing Summary - Design Initialization	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Opt Design (opt_design)				
DRC - Opt Design	report_drc		6/23/26, 10:25 AM	4.6 KB
Timing Summary - Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Power Opt Design (power_opt_design)				
Timing Summary - Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Place Design (place_design)				
IO - Place Design	report_io		6/23/26, 10:25 AM	205.9 KB
Utilization - Place Design	report_utilization		6/23/26, 10:25 AM	10.5 KB
Control Sets - Place Design	report_control_sets	verbose = true;	6/23/26, 10:25 AM	3.5 KB
Incremental Reuse - Place Design	report_incremental_reuse			
Incremental Reuse - Place Design	report_incremental_reuse			
Timing Summary - Place Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Post-Place Power Opt Design (post_place_power_opt_design)				
Timing Summary - Post-Place Power Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Post-Place Phys Opt Design (phys_opt_design)				
Timing Summary - Post-Place Phys Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true;		
▼ Route Design (route_design)				
DRC - Route Design	report_drc		6/23/26, 10:25 AM	4.6 KB
Methodology - Route Design	report_methodology		6/23/26, 10:25 AM	1.3 KB
Power - Route Design	report_power		6/23/26, 10:25 AM	8.1 KB
Route Status - Route Design	report_route_status		6/23/26, 10:25 AM	0.6 KB
Timing Summary - Route Design	report_timing_summary	max_paths = 10; routable_nets = true; report_unconstrained = true;	6/23/26, 10:25 AM	48.4 KB
Incremental Reuse - Route Design	report_incremental_reuse			
Clock Utilization - Route Design	report_clock_utilization		6/23/26, 10:25 AM	7.1 KB
Bus Skew - Route Design	report_bus_skew	warn_on_violation = true;	6/23/26, 10:25 AM	0.9 KB
implementation_log			6/23/26, 10:25 AM	31.4 KB
▼ Post-Route Phys Opt Design (post_route_phys_opt_design)				
Timing Summary - Post-Route Phys Opt Design	report_timing_summary	max_paths = 10; report_unconstrained = true; warn_on_violation = true;		
Bus Skew - Post-Route Phys Opt Design	report_bus_skew	warn_on_violation = true;		
▼ Write Bitstream (write_bitstream)				
implementation_log				

Synthesis: 6/9/26 11:02 AM | Implementation (Place+Route): 6/23/26 10:25 AM | All design stages completed successfully





Conclusion & Future Scope

Achievements

- Successfully designed 8-bit ALU with 8 operations
- Fully verified through Vivado simulation (all 5 test cases PASS)
- RTL schematic, synthesis & implementation completed
- Power analysis and DRC reports generated
- Purely combinational design using always @(*)

Future Enhancements

- Add signed arithmetic (overflow detection)
- Implement multiply and divide operations
- Add XDC constraints for bitstream generation
- FPGA deployment on Basys3 / Nexys board
- Expand to 16-bit or 32-bit ALU for RISC core

"This project establishes a strong foundation for digital design, processor architecture, and FPGA implementation."



Thank You

Aryan Singh

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VLSI Domain — Minor Project

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